Age

15 - 17

17 or below - 1

18 - 21 - 2

22 - 25 - 3

26 - 29 - 4

30 or above - 5

30 - 40 $s=$

~~Ans~~

$\bar{x} = 21.90845$

$\bar{x} \approx 21.91$ years

$$\bar{x} - t_{\alpha/2, v} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2, v} \times \frac{s}{\sqrt{n}}$$

~~21.203 < μ < 22.613~~

0.95

1.971

$\mu < 21.91 + 1.971 \times$

$\Sigma\, age^2$ $\Sigma\, age = 4666.5$

108011.25

5.219

sd

$$108011.25 - \frac{(4666.5)^2}{213}$$

$$\frac{41 0058}{71} = 27.24278844$$

Inference of mean age of population

mean age $= 21.90845$
$= 21.908$

$$\bar{x} - t_{\alpha/2, v} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2, v} \frac{s}{\sqrt{2}}$$

$n = 213$, 95 % confidence interval

$s = 5.219$

$t_{\alpha/2, v} = 1.971$

$21.203 < \mu < 22.613$

HYPOTHESIS TEST FOR MEAN AGE (0.05 significance level)

$H_0 : \mu = 25$, $df = n-1 = 213-1 = 212$
$H_1 : \mu \neq 25$

$t_{cal} = \dfrac{21.908 - 25}{s/\sqrt{n}}$

$= \dfrac{21.908 - 25}{5.219/\sqrt{213}}$, $df = 212$

$= -8.6465$

$t_{crit} = \pm 1.971 = t_{\alpha/2, 212}$

$-8.6465 < -1.971$,

reject H_0

refer to chi-squared 1 folder.

Chi-squared (at 0.05 level)

Gender / Personal goals	Yes	NO
MALE	76	16
FEMALE	107	14

H_0: There is no association b/w gender & making personal goals to achieve a target

H_1: There is an association between gender & making personal goals to achieve it.

Chi-squared stat = 1.0219

Critical value to reject: 3.84145

if chi-squared stat > critical value:
  reject Ho

else accept.

∴ fail to reject Ho.
  $1.0219 < 3.84145$

To check between gender & having a clear goal in life

chi-squred 2

Gender / Clear goal	YES	NO
MALE	59	33
FEMAE E	88	33

H_0: There is no association b/w gender & having a clear goal in life

H_1 : H_0 is false

Chi-squared stat = 1.42657

dof = (2-1)(2-1) = 1

1.42657 < 3.84145

fail to reject null ~~hypothe~~ hypothesis

(α = 0.05)

chisquared - 3 folder ↳ refer

chi sq test

Family system & being worried about the future

FAMILY SYSTEM / WORRY	YES	NO
Nuclear	60	71
Joint	28	35
Living Alone	7	12

$df = (3-1)(2-1) = 2$

chi_stat = 0.539959

$df = 2$

crit = 5.991

$0.539959 < 5.991$,

fail to reject H_0.

Mode for family system = Nuclear

Number of siblings

Ranks	Actual	Counts
1	0	6
2	1	33
3	2	48
4	3	58
5	4	68

mode = 4 siblings
median = 3 siblings

Hypothasis
Avg no of siblings is 3

Test

H_0 : $\mu = 3$
H_1 $\mu < 3$

$\bar{x} = 2.6995$
$sd = 1.1549$

$$t_{cal} = \frac{2.6995 - 3}{\frac{2.6995}{\sqrt{213}}} = -1.6246$$

$t_{crit} = -1.652073$

$t_{cal} > t_{crit}$ ∴ fail to reject H_0

Testing porpotion of people who procrastinate often & play sports often vs. people who procrastinate often and play sports rarely.

Test if $p_1 < p_2$, $\alpha = 0.05$

n_1 = play sport often sample
n_2 = play sport rarely sample

$n_1 = 41$
$n_2 = 42$

in n_1 13 people procrastinate often so $\hat{p}_1 = \frac{13}{41}$

in n_2 19 people procrastinate often so $\hat{p}_2 = \frac{19}{42}$

$H_0 : p_1 = p_2$
$H_1 : p_1 < p_2$

$$Z_{cal} = \frac{\left(\frac{13}{41}\right) - \left(\frac{19}{42}\right) - 0}{\sqrt{p_c\, q_c \left(\frac{1}{41} + \frac{1}{42}\right)}}$$

$p_c = \frac{32}{83}$, $q_c = \frac{51}{83}$

$$\frac{\left(\frac{13}{41}\right) - \left(\frac{19}{42}\right) - 0}{\sqrt{\frac{32}{83} \times \frac{51}{83} \times \left(\frac{1}{41} + \frac{1}{42}\right)}}$$

$z_{cal} = \cancel{41206}\ -1.19$

$z_\alpha = -1.645$

$\cancel{z_{cal} < z_\alpha}$

$z_{cal} > z_\alpha$, fail to reject H_0.

Spearman Correlation between level of confidence & frequency of procrastination.

mode of level of confidence in people $=2=$ confident

mode of procrastination level $=2$ $=$ often

refer python code

Correlation coefficient:

$$-1 \leq r_s \leq 1$$

if tend to -1, negative correlation,

if tent to $+1$, positive correlation

if close to 0, little association between the two variables

$r_s \approx -0.04136$, $p=0.5265$

Conclusion : r_s is very close to 0 & p value is more than 0.05, it shows a very weak correlation b/w the two variables, Ho cannot be rejected i.e there is no correlation b/w the two variables, more samples need to be collected to reach a conclusion

Test hypothesis

p = porpotion of people who procastinate ~~Often~~ Always

~~[...]~~

$\hat{p} = 97 / 213$

~~Fin~~ Finding interval for p

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{pq}{n}} < p < \hat{p} + z_{\alpha/2}\sqrt{\frac{pq}{n}}$$

$$\cancel{\frac{97}{213} - 1.96\sqrt{(97/213)(116/213)} < p < \frac{97}{213} + 1.96\sqrt{\frac{(97/213)(116/213)}{213}}}$$

$$\frac{97}{213} - 1.96\sqrt{\frac{(97/213)(116/213)}{213}} < p < \frac{97}{213} + 1.96\sqrt{\frac{(97/213)(116/213)}{213}}$$

$$0.3885 < p < 0.5223$$

Test hypothesis

~~$H_0 : p = 0$~~
~~$H_1 : p \neq$~~

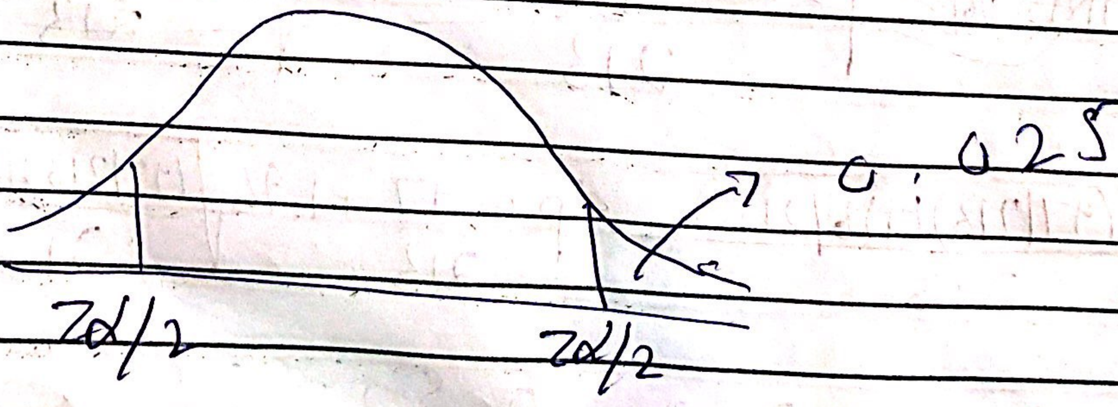
~~$H_0 = f$~~

$H_0 : p = 0.5$
$H_1 : p \neq 0.5$

$\alpha = 0.05$

$$\frac{(97/213) - 0.5}{\sqrt{\dfrac{0.5 \times 0.5}{213}}}$$

$z_{cal} = -1.302$



$-1.96 < -1.302 < 1.96$

$\therefore$ fail to reject H_0

Interval for difference b/w porportions

~~Proportion of people who play sports often to procrastinate~~ ~~vs~~ often

n_1 = no. of people who play sports often

n_2 = no. of people who play sports rarely

p_1 = people who procastinate often among ~~the~~ ~~sample~~ those who play sports often

p_2 = people who procrastinate often among those who play sports rarely

$n_1 = 41$

$n_2 = 42$

$\hat{p}_1 = 13/41$

$\hat{p}_2 = 19/42$

$$\text{lower} = (\hat{p}_1 - \hat{p}_2) - z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$\text{upper} = (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$\text{lower} = \left(\frac{13}{41} - \frac{19}{42}\right) - 1.96\sqrt{\frac{(13/41)(28/41)}{41} + \frac{(19/42)(23/42)}{42}}$$

$$\text{lower} = -0.3425$$

$$\text{upper} = 0.0719$$

$$-0.3425 < p_1 - p_2 < 0.0719$$

Two demographics
Nuclear vs joint household

Procrastinating often

p_1 = porpotion of ~~people~~ population procrastinate often in nucleas

p_2 = porpotion of population procrastinate often in joint

$\hat{p}_1 = 50/131$, $n_1 = 131$, $n_2 = 63$

$\hat{p}_2 = \cancel{63/123}\ 21/63$

$H_0: p_1 = p_2$
$H_1: p_1 \neq p_2$

$\cancel{z_{cal} = \left(\frac{50}{131}\right), \left(\frac{63}{131}\right)}$

$p_c = \frac{71}{194}$

$q_c = \frac{123}{194}$

$$z_{cal} = \frac{\left(\frac{50}{131}\right) - \left(\frac{21}{63}\right)}{\sqrt{\frac{71}{194} \times \frac{123}{194}\left(\frac{1}{131} + \frac{1}{63}\right)}} = 0.6546$$

$\alpha = 0.05$

$Z_{\alpha/2} = \pm 1.96$

$-1.96 < 0.6546 < 1.96$

fail to reject H_0.

* refer graph

Hypothesis test (90% confidence interval)

~~completely depen~~

Somewhat dependent $(n_1) = 83$

Neither dependent or independent $= 55$

p_1 = somewhat dependent & procastinate often porpotion

p_2 = neither dependent nor independent (neutral) & procrastinate often porportion

$H_0 : p_1 = p_2$

$H_1 : p_1 \neq p_2$

$\hat{p}_1 = 38/83$, $\hat{p}_2 = 15/55$

$$p_c = \frac{38+15}{83+55} = \frac{53}{138}$$

$$q_c = \frac{85}{138}$$

$$z_{cal} = \frac{\left(\frac{38}{83}\right) - \left(\frac{15}{55}\right) - 0}{\sqrt{\frac{53}{138} \times \frac{85}{138} \times \left(\frac{1}{83} + \frac{1}{55}\right)}} = 1.749$$

$z_{\alpha/2} = 1.645$

$|Z_{cal}| > z_{\alpha/2}$

reject H_0 at 90% confidence level